MIL-C-19729A (WEP)
12 May 1964
Superseding
MIL-C-19729 (AER)
28 September 1956

MILITARY SPECIFICATION

COMPUTER, AERIAL PHOTOGRAPHIC DATA

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 SCOPE. - This specification covers the requirements for a manual, pocket size, concentric disc computer for calculating aerial photographic and multi-sensor data. The computer is designated as Type BM-38 ( ). See 6.3.

2. APPLICABLE DOCUMENTS

2.1 SPECIFICATIONS AND STANDARDS. - The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

L-V-351	Vinyl Chloride Polymer and Copolymer Rigid Sheets
LLL-B-631	Boxes, Fiber, Corrugated (For Domestic Shipment)
LLL-B-636	Boxes, Fiber, Solid (For Domestic Shipment)
PPP-B-601	Boxes, Wood, Cleated, Plywood

Military

JAN-P-106	Packaging and Packing for Overseas Shipment: Boxes, Wood, Nailed
MIL-P-116	Preservation, Method of

FSC 6760

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STANDARDS

Federal

FED-STD-102	Preservation, Packaging and Packing Levels

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property

(When requesting specifications and other publications, refer to both title and number. Copies of applicable specifications and standards may be obtained upon application to the Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pennsylvania, 19120, Attn: Code DCI)

3. REQUIREMENTS

3.1 DESCRIPTION. - The computer shall consist of four circular discs and a cursor pivoted and fastened together concentrically. The face (obverse) side shall have scales, indices, and information that pertain to aerial photography. The other (reverse) side shall have scales, indices and information that pertain to electronic sensors.

3.2 DATA. - No data is required by this specification, or by applicable documents referenced in Section 2, unless specified in the contract or order. (See 6.2).

3.3 DESIGN AND CONSTRUCTION. - The contractor shall be responsible for the detailed and final design of the computer and its compliance with the requirements of this specification. When required in the procurement document (see 6.2) and prior to fabrication, the contractor shall submit to the contracting officer preliminary sketches and drawings of the proposed design for approval by the Photographic Management Division, Bureau of Naval Weapons. Approval of the drawings shall not relieve the contractor of his responsibility to fulfill the requirements of this specification. Fabrication of any computer prior to approval of the preliminary drawings shall be at the contractor's risk.

3.3.1 Obverse Side. - The face of the computer shall be provided with scales and indices, so positioned and related to one another that when altitude, focal length, and ground speed are given, the operator can rapidly and accurately read the following information with one setting of the computer:

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a. Ground to image scale, as a scale factor.

b. Ground covered along line of flight in linear feet for 2-1/4 x 2-1/4, 4-1/2 x 4-1/2, and 9 x 9 inch format photographs.

c. Ground gained forward in linear feet at 60 percent forward overlap for 2-1/4 x 2-1/4, 4-1/2 x 4-1/2, and 9 x 9 inch format photographs.

d. Ground gained forward in linear feet at 66 percent forward overlap for 4-1/2 x 4-1/2 inch format photographs.

e. Scale in linear feet on ground per inch on negative.

f. Exposure interval in seconds at 60 percent forward overlap for 2-1/4 x 2-1/4, 4-1/2 x 4-1/2, and 9 x 9 inch format photographs.

g. Exposure interval in seconds at 66 percent forward overlap for 4-1/2 x 4-1/2 inch format photographs.

h. Ground distances in linear feet corresponding to negative distances in decimal fractions of a foot and to resolution in lines per millimeter (ll/mm) on the negative.

i. Distance to optical horizon in nautical miles (nm).

3.3.2 Reverse Side. - The reverse side of the computer shall be provided with scales and indices so positioned and related to one another that:

a. When format width and scan angle of an infra-red (IR) set are given, the operator can determine the effective focal length (EFL) of the system.

b. When format width, scan angle, image distance from nadir to an object, and altitude are given, the operator can read the angle subtended from the nadir to the object, can read the tangent of this angle and can determine the ground distance with respect to the nadir.

c. When EFL and altitude are given, photo scale at nadir can be determined.

3.4 PREPRODUCTION SAMPLE. - Unless otherwise specified (see 6.2) six computers shall be manufactured using the same methods and procedures as will be used for quantity production and shall be submitted as the preproduction sample for acceptance. Approval of the preproduction sample by the requiring activity shall not relieve the contractor of his obligation to supply equipment

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conforming to this specification. The preproduction sample shall be applied as part of the quantity specified by the contract or order.

3.5 MATERIAL. - The material used shall be of good commercial quality free from all defects and imperfections that may affect serviceability of the finished computer. The material shall be dimensionally stable, shall not warp, and shall meet the requirements of this specification.

3.5.1 Plastic. - The discs and cursor shall be vinyl resin conforming to Federal Specification L-V-351.

3.5.2 Metal. - All metal used in fabrication of the computer shall be corrosion resistant.

3.6 DIMENSIONS. - The maximum diameter of the computer shall be 5.3 inches. The computer shall be as thin as possible, compatible with its intended use.

3.7 MARKING AND LETTERING. - The computer shall be inscribed with black marks, figures, and letters on a white background. Except as specified herein, all lettering and figures shall conform to the curvature of the computer and shall be Heavy Plate Gothic. Where space is limited, scale and index titles may be abbreviated. The markings shall be so made that they will not smear or wear off in use. The width of the scale lines shall be approximately 0.01 inch.

3.8 ACCURACY. - The accuracy of the computer shall be such that the alignment of any index with related scale graduations, or the alignment between graduations of related scales shall not deviate by more than plus or minus 0.01 inch from the true value.

3.9 ENVIRONMENTAL CONDITIONS

3.9.1 Operating Environment Conditions. - The computer shall operate as specified in any ambient temperature from 0 degrees to 100 degrees F. at relative humidities up to 90 percent and shall pass the performance, accuracy and dimensional stability tests.

3.9.2 Storage Environment Conditions. - The computer shall not be damaged by storage at ambient temperatures ranging from 180 degrees to minus 65 degrees F. at relative humidities up to 90 percent, and shall pass the performance, accuracy and dimensional stability tests.

3.10 PERFORMANCE. - The discs of the computer shall be fastened snugly together and shall be capable of being set easily

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and rapidly without the use of undue force. The accuracy of the settings shall be as specified herein. The legibility of the markings shall provide easy and accurate readability. The computer shall pass the performance test given in Section 4.

3.11 COMPONENTS. - The computer shall consist of a base disc, three complimentary discs and a cursor. The complimentary discs and cursor shall be fastened to the base disc and shall pivot concentrically with the base disc. One complimentary disc, designated the photographic disc, shall be on the obverse of the base disc. Two complimentary discs, designated as inner and outer discs, and the cursor shall be on the reverse of the base disc. The base disc and the complimentary discs shall be marked with scales and indices as specified herein. The complimentary discs and cursor shall be provided with clear areas as needed, to read underlying scales.

3.12 OBVERSE SIDE SCALES. - The scales to be placed on the obverse of the computer and their upper and lower limits of extension shall be as listed below. All the scales shall be subdivided as specified herein and in the smallest increments possible while maintaining clarity and accuracy. The scales shall be laid out along concentric arcs. Starting at the periphery and approaching the center, the scales shall be placed in the following order.

3.12.1 Base Disc (Obverse)

3.12.1.1 Feet on Ground Scale. - The "feet on ground" scale shall be calibrated in feet and shall extend counter-clockwise from 0.1 to at least 100, and shall be labeled FEET ON GROUND (Resolution = Bar + Space).

3.12.1.2 Scale Factor Scale. - The "scale factor" scale shall extend counter-clockwise at least from 250 to 80,000 and shall be labeled SCALE FACTOR.

3.12.1.3 Focal Length Scale. - The "focal length" scale shall be calibrated in inches and shall extend clockwise from 0.5 to 200. The following focal lengths shall be marked clearly in bold type: 1-1/2, 1-3/4, 3, 100mm, 6, 7, 12, 18, 20, 24, 36, 48, 96, 100, and 200 inches. The scale shall be labeled FOCAL LENGTH.

3.12.1.4 Exposure Interval Scales. - Separate scales shall be provided as needed for computing exposure interval in seconds for 60 percent forward overlap on 2-1/4 x 2-1/4, 4-1/2 x 4-1/2, and 9 x 9 inch format photographs and 66 percent forward overlap for 4-1/2 x 4-1/2 inch format photographs. The exposure interval scale shall extend counter-clockwise from 0.025 to 80 seconds for the 2-1/4 x 2-1/4 and 4-1/2 x 4-1/2 inch format photographs. The exposure interval scale for 9 x 9 inch format photographs shall

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extend counter-clockwise from 0.05 to 80 seconds. The scales shall be labeled EXPOSURE INTERVAL.

3.12.1.5 Ground Coverage Scale. - The "ground coverage" scale shall be calibrated in feet, shall extend counter-clockwise from 50 to 100,000 and shall be labeled GROUND COVERAGE.

3.12.2 Photographic Disc

3.12.2.1 Feet on Negative Scale. - The "feet on negative" scale shall be calibrated in feet, shall extend counter-clockwise from 0.00005 to 0.1 feet and shall be labeled FEET ON NEGATIVE.

3.12.2.2 Image Resolution Scale. - The "image resolution" scale shall be calibrated in lines per millimeter, shall extend clockwise from 10 to 200 and shall be labeled IMAGE RESOLUTION, LINES PER MILLIMETER.

3.12.2.3 Altitude Scale. - The "altitude" scale shall be calibrated in feet, shall extend clockwise from 250 to 200,000 and shall be labeled ALTITUDE.

3.12.2.4 Distance to Horizon (nm) Scale. - The "distance to horizon (nm)" scale shall be calibrated in nautical miles, the increments shall extend clockwise in direct relationship to the altitude scale increments, and shall be labeled DISTANCE TO HORIZON (nm).

3.12.2.5 Ground Speed Scale. - The "ground speed" scale shall be calibrated in knots, shall extend clockwise from 100 to 3,000, and shall be labeled GROUND SPEED/KNOTS.

3.13 OBVERSE SIDE INDICES. - Clearly legible index marks shall be placed on the obverse of the computer as listed below. Each index mark shall be clearly visible and shall be labeled according to its use.

3.13.1 Scale Index. - An arrow shall be placed so that it will indicate the correct scale factor, within the limits of that scale, for any combination of altitude and focal length on the computer and shall be labeled SCALE INDEX.

3.13.2 Ground Coverage Indices. - Arrows shall be placed so that the ground covered along the line of flight by 2-1/4 x 2-1/4, 4-1/2 x 4-1/2, and 9 x 9 inch format photographs can be read on the "ground coverage" scale, within the limits of that scale, and appropriately labeled.

3.13.3 Ground Gained Indices. - Arrows shall be placed so that the ground gained forward at 60 percent forward overlap for

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2-1/4 x 2-1/4, 4-1/2 x 4-1/2, and 9 x 9 inch format photographs and for 66 percent forward overlap for 4-1/2 x 4-1/2 inch format photographs can be read on the "ground coverage" scale, within the limits of that scale, and appropriately labeled.

3.13.4 Scale in Feet per Inch on Negative Index. - An arrow shall be placed so that it will indicate on the "ground coverage" scale the number of feet on the ground per inch on the negative, within the limits of that scale, and appropriately labeled.

3.14 REVERSE SIDE SCALES. - The scales to be placed on the reverse of the computer, and their upper and lower limits of extension shall be as listed below. All scales shall be subdivided decimally in the smallest increments possible while maintaining clarity and accuracy. The scales shall be laid out along concentric arcs. Starting at the periphery and approaching the center, the scales shall be placed in the following order. The degrees of arc indicated are approximate.

3.14.1 Base Disc (Reverse)

3.14.1.1 Image or Format Size Scale. - The "image or format size" scale shall be calibrated in inches and shall extend from 0.01 to 36, logarithmically, in a clockwise direction. The scale shall consist of four segments; 0.01 to 0.1, 0.1 to 1.0, 1.0 to 10, each segment covering a 100 degree arc, and a fourth segment extending from 10 to 36, over a 57 degree arc. The scale shall be labeled IMAGE OR FORMAT SIZE.

3.14.2 Inner Disc

3.14.2.1 Scan Angle Degrees Scale. - The "scan angle degrees" scale shall extend from 35 minutes to 180 degrees in a clockwise direction and shall be so oriented and spaced that when "pi" on the "image or format size" scale is set opposite 180 degrees on the "scan angle degrees" scale, the image scale will give the radians equivalent to each angle. The scale will cover an arc of 255 degrees and shall be labeled SCAN ANGLE DEGREES.

3.14.2.2 Tangent Scale. - The "tangent scale" shall extend from 0.01 to 9 in a clockwise direction and shall be so oriented and spaced that 45 degrees on the "scan angle degrees" scale will be positioned opposite 1.0 on the tangent scale. This scale shall cover an arc of 215 degrees and shall be labeled TANGENT.

3.14.3 Outer Disc

3.14.3.1 Altitude or Ground Scale. - The "altitude or ground"

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scale shall be calibrated in feet, shall extend from 100 to 100,000 in a clockwise direction over an arc of 400 degrees (a complete circle plus a 40 degree overlap) and shall be labeled ALTITUDE OR GROUND SCALE.

3.15 REVERSE SIDE INDICES

3.15.1 Image or Format Size Scale Mark. - A line shall be placed on the "image or format size" scale located at the 3.1416 increment. The line shall be labeled with the "pi" symbol.

3.15.2 Scan Angle Degree Scale Mark. - A line shall be placed on the "scan angle degree" scale located at the 57.3 increment. The line shall extend to the outer edge of the inner disc and shall be labeled E.F.L.

3.16 TABLE OF SLANT RANGE FACTORS. - A compact, legible table giving slant range factors commonly used in oblique photography shall be placed near the center of the photographic disc. This table need not conform to the curvature of the computer. The values to be included shall be as follow:

Slant range factor	20	11	8.27	4	3.04	2
Depression angle, degrees	3	5	7.25	15	19.75	30

3.17 SIDE LOOKING RADAR (SLR) FACTOR TABLE. - A compact, legible table, giving side looking radar (SLR) factors for the mode of operation, shall be placed near the center of the outer disc of the reverse of the computer. The table shall be as follows and labeled SLR.

SLR

Mode	Scale Factor
1	109,370
2	218,740
3	729,134
4-7	364,567

3.18 INSTRUCTIONS. - Instructions for setting the computer to solve problems shall be marked on the obverse and reverse sides of the computer. The printing may be placed near the center of each disc and need not conform to the curvature of the computer.

3.18.1 Obverse Instructions. - The instructions for setting the obverse of the computer shall be placed on the photographic disc and shall be as follows:

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INSTRUCTIONS

SET ALTITUDE OPPOSITE LENS FOCAL LENGTH OR
SET SCALE INDEX AT REQUIRED SCALE FACTOR

3.18.2 Reverse Instructions. - The instructions for setting the reverse of the computer shall be placed on the outer disc and shall be as follows:

INSTRUCTIONS

SET IMAGE OR FORMAT SIZE TO SCAN ANGLE DEGREES AND

a. EFL (Effective Focal Length) is obtained from Image or Format Size Scale at EFL index.

b. GD (Ground Distance referenced to Nadir) is obtained by setting cursor at image distance on Image or Format Size Scale. With cursor read Tangent Scale. Multiply tangent by altitude.

3.18.3 SLR Table Instructions. - The instructions for using the SLR Table shall be placed directly under the SLR Table. The instructions shall be as follow:

Multiply the SLR Scale Factor, for the mode used, by the Image or Format Size.

3.19 IDENTIFICATION MARKING. - Identification marking shall be in accordance with Military Standard MIL-STD-130.

3.20 OTHER MARKINGS. - No scales or information shall be placed on the computer except as specified herein.

3.21 ENVELOPE. - Each computer shall be provided with a suitable protective plastic envelope which may be marked with detailed instructions for operating the computer and shall be marked with the specified identification marking. The envelope shall not be damaged when subjected to the storage environment test as specified herein.

3.22 OPERATING INSTRUCTIONS. - When specified in the contract or order (see 6.2), additional operating instructions shall accompany the computer.

3.23 WORKMANSHIP. - Workmanship shall be of the highest grade throughout and in accordance with good commercial practices for this type of equipment. The computer shall be neat in appearance and free from burrs and sharp edges.

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4. QUALITY ASSURANCE PROVISIONS

4.1 RESPONSIBILITY FOR INSPECTION. - Unless otherwise specified in the contract or purchase order (see 6.2), the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified (see 6.2), the supplier may utilize his own or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 TEST CONDITIONS AND MATERIEL. - Unless otherwise specified herein, tests shall be conducted under ordinary room conditions of temperature and humidity. The equipment and supplies necessary for testing shall be furnished by the contractor.

4.3 CLASSIFICATION OF INSPECTIONS. - Inspections shall be as listed in Table I and shall be classified as follows:

(a) Preproduction Inspection
(b) Quality Conformance Inspection
(c) Preparation for Delivery Inspection

TABLE I
EXAMINATIONS AND TESTS

Test	Requirement Paragraph	Test Method
PREPRODUCTION INSPECTION		
Visual and Mechanical Examination	3.3, 3.5, 3.6 3.7, 3.10, 3.11, 3.12, 3.16, 3.17, 3.18, 3.19, 3.20, 3.21, 3.23	4.10.1
Performance Test	3.3.1, 3.3.2	4.10.2
Operating Environmental Test	3.9.1	4.10.3
Storage Environmental Test	3.9.2	4.10.4
Warpage Test	3.5	4.10.5
Dimensional Stability Test	3.5	4.10.6
Accuracy Test	3.8	4.10.7

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TABLE I (Cont'd.)

EXAMINATIONS AND TESTS

TEST	Requirement Paragraph	Test Method
QUALITY CONFORMANCE INSPECTION		
Visual and Mechanical Examination	3.3, 3.5, 3.6, 3.7, 3.10, 3.11, 3.12, 3.16, 3.17, 3.18, 3.19, 3.20, 3.21, 3.23	4.10.1
Performance Test	3.3.1, 3.3.2	4.10.2
Accuracy Test	3.8	4.10.7
PREPARATION FOR DELIVERY INSPECTION		
Preservation	5.1	4.6
Packaging	5.1	4.6
Packing	5.2	4.6
Marking	5.3	4.6

4.4 PREPRODUCTION INSPECTION. - Prior to submission for acceptance, the preproduction sample shall be subjected by the contractor to the preproduction inspection specified herein. On completion of the inspection by the contractor, the preproduction sample shall be shipped to the Military Testing Agency designated by the requiring activity (see 6.2). Unless otherwise specified, (see 6.2), a report of the results of the tests shall accompany the preproduction sample. The Military Testing Agency may repeat the preproduction inspection or make other tests as needed to verify that the production methods and choice of design details are capable of yielding equipment that complies with the requirements of this specification. Approval by the requiring activity shall be based on no defects in the preproduction sample.

4.5 QUALITY CONFORMANCE INSPECTION. - Each production computer, subsequent to the preproduction sample, submitted for acceptance under the contract, or order, shall be subjected by the contractor to the quality conformance inspection specified herein. The quality conformance inspection shall verify that all subsequent computers are duplicates of the approved preproduction sample.

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4.6 PREPARATION FOR DELIVERY INSPECTION. - Examine the preservation, packaging, packing and marking for shipment and storage, to determine conformance to the requirements of the applicable specifications referenced herein.

4.7 LOT. - For the purposes of sampling, a lot shall consist of all the computers offered for acceptance at one time.

4.8 SAMPLING. - Sampling for all inspections shall be in accordance with Military Standard MIL-STD-105. The inspection level shall be General Inspection Level II and the acceptance quality level (AQL) shall be 1.0 percent defective for all defects.

4.9 CLASSIFICATION OF DEFECTS. - Classification of defects shall be as defined in Military Standard MIL-STD-105.

CRITICAL DEFECTS:

1. Markings illegible, omitted, blurred, smudged, etc.
2. Discs or cursor missing.
3. Disc movement not concentric

MAJOR DEFECTS:

101. Discs do not run easily.
102. Discs too loose.
103. Computer not clean.
104. Dimensions not as specified.
105. Color not as specified.
106. Material not as specified.

MINOR DEFECTS:

None defined.

4.10 TESTS AND EXAMINATIONS

4.10.1 Visual and Mechanical Inspection. - Inspect the computer visually and mechanically to determine that it is a duplicate of the preproduction sample and with respect to completeness, cleanliness, dimensions, performance, accuracy, and workmanship.

4.10.2 Performance Test. - Rotate the discs of the computer to ascertain easy and positive movement of the discs. Set the discs, obverse and reverse, on any one setting and drop the computer from a height of 30 inches three times. Check the setting. The setting shall not have been dislodged more than 1/16 inch. The computer shall not have been damaged. Failure to pass the performance test shall be cause for rejection.

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4.10.3 Operating Environment Test. - Place the computer for four hours in a test chamber with the temperature lowered to 0 F. At the end of the four hour period, with the temperature at 0 F, conduct the performance, accuracy and dimensional stability tests. Raise the temperature in the test chamber to 100 F. and increase the humidity to 90 percent. At the end of the four hour period, with the temperature at 100 F, conduct the performance, accuracy and dimensional stability tests. Failure of the computer to pass the performance, accuracy and dimensional stability tests shall be cause for rejection.

4.10.4 Storage Environment Test. - Place the computer and the computer's protective envelope in a test chamber and lower the temperature to minus 65 F. Over a twenty-four period, cycle the humidity at least once to include a combination which shall cause moisture to condense on the computer. At the end of the 24 hour period, raise the temperature to 160 F. and raise the humidity from 10 percent to 90 percent over a 24 hour period. After the computer and envelope have returned to room temperature, they shall be carefully examined for signs of damage. The computer shall be subjected to the performance, accuracy, and dimensional stability tests. Signs of damage to the computer and envelope and failure of the computer to pass the performance, accuracy and dimensional stability tests shall be cause for rejection of the computer or envelope whichever does not pass the test.

4.10.5 Warpage Test. - The computer shall be placed in a test chamber with one edge of the computer resting on a wooden block one inch high and the temperature raised to 160 F. Raise the humidity from 10 percent to 90 percent over a period of twenty-four hours. At the end of this period, inspection of the computer shall be made to determine that no warpage or other deleterious effect has occurred. This test may be performed simultaneously with the storage environment test. Failure to pass this test shall be cause for rejection.

4.10.6 Dimensional Stability Test. - To meet the dimensional stability requirements, the computer shall pass the accuracy tests specified in the storage environment, operating environment, and warpage tests.

4.10.7 Accuracy Test. - To pass the accuracy test, the markings shall align within the accuracy tolerance specified herein. The accuracy of the markings of the obverse and reverse sides of the computer shall be tested as indicated below.

4.10.7.1 Obverse Side Accuracy Test. - Set the focal length scale increment marked 12 inches opposite the altitude scale increment marked 10000. The scales should read as indicated below:

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a. Focal length of 3 inches should align with 2500 foot altitude.

b. Focal length of 36 inches should align with 30000 foot altitude.

c. Focal length of 96 inches should align with 80000 foot altitude.

d. Scale index should align with scale factor of 10000.

e. One foot on feet on ground scale should align with 32.8 lines/mm on the image resolution scale.

f. One foot on feet on ground scale should align with 0.0001 foot on feet on negative scale.

g. 9 x 9 inch negative ground coverage index should align with 7500 feet.

h. 4-1/2 x 4-1/2 inch negative ground coverage index should align with 3750 feet.

i. 9 x 9 inch negative ground gained at 60 percent overlap index should align with 3000 feet.

j. 2-1/4 x 2-1/4 inch negative ground coverage index should align with 1875 feet.

k. 4-1/2 x 4-1/2 inch negative ground gained at 60 percent overlap index should align with 1500 feet.

l. 4-1/2 x 4-1/2 inch negative ground gained at 66 percent overlap index should align with 1275 feet.

m. Feet per inch on negative scale index should align with 833 feet.

n. 2-1/4 x 2-1/4 inch negative ground gained at 60 percent overlap index should align with 750 feet.

o. Ground speed of 900 knots increment for 4-1/2 x 4-1/2 inch format at 60 percent overlap should align with exposure interval of 1.0 second.

p. Ground speed of 900 knots increment for 2-1/4 x 2-1/4 inch format at 60 percent overlap should align with exposure interval of 0.5 second.

q. Ground speed of 900 knots increment for 9 x 9 inch format at 60 percent overlap should align with exposure interval of 2.0 second.

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4.10.7.2 Reverse Side Accuracy Test. - Set the E.F.L. index on the inner disc to align with 0.1 increment of the image or format size scale. With the cursor, align the 1000 foot increment of the altitude scale to the same mark. The scales on the reverse should read as follows:

Image or Format Size Scale	1	2	.2	.1
Scan Angle Degrees Scale	57.3	115	11.5	5.8
Tangent Scale	1.573	n/a	2	.1
Altitude or Ground Stale	1000	2000	800000	400000

5. PREPARATION FOR DELIVERY

5.1 PRESERVATION AND PACKAGING. - Preservation and packaging shall be Level A as defined in Federal Standard No. 102. The computer in its protective envelope shall be packaged as one unit.

5.1.1 Cleaning. - The computer shall be cleaned following the procedures specified in Specification MIL-P-116, for C-1 cleaning.

5.1.2 Drying. - Cleaned surfaces shall be dried in accordance with procedures specified in D-4, wiping as prescribed in Specification MIL-P-116.

5.1.3 Preservation. - Preservatives shall not be used.

5.1.4 Unit Package. - The computer shall be packaged by Method III as prescribed in Specification MIL-P-116.

5.1.5 Intermediate Package. - Unit packages shall be packaged in intermediate containers conforming to Specification LLL-B-631 or LLL-B-636. Intermediate containers shall contain no more than 25 unit packages.

5.2 PACKING. - The level of packing shall be as specified by the procuring activity (see 6.2).

5.2.1 Level A. - Intermediate containers shall be packed for shipment in exterior containers conforming to Specification PPP-B-601 or JAN-P-106.

5.2.2 Level C. - Intermediate packages that require overpacking for acceptance by the carrier shall be packed in exterior type shipping containers that will insure safe transportation at the lowest rate to the nearest point of delivery.

5.3 MARKING. - All unit packages, intermediate containers and shipping containers shall be marked in accordance with Military Standard MIL-STD-129.

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6. NOTES

6.1 INTENDED USE. - The computer is intended for making calculations used in the planning of aerial photographic flights and in analysis of post-flight images.

6.2 ORDERING DATA. - Procurement documents should specify the following:

a. Title, number and date of this specification

b. Quantity required

c. Sketches and drawings, when required (see 3.3)

d. Preproduction samples, when required (see 3.4)

e. Operating instructions, when required (see 3.22)

f. Responsibility for inspection (see 4.1)

g. Preproduction test report, if not required (see 4.4)

h. Designation of Military Testing Agency (see 4.4)

i. Selection of packing level (see 5.2)

6.3 TYPE DESIGNATOR. - In accordance with the provisions of Standard MIL-STD-105, the parenthesis in the type designation will be replaced by a letter, number or both. This information will be furnished upon application by the contractor to the Requiring Activity after award of each contract. The complete type designation shall then be used on nameplates, in instruction books and wherever reference is made to this specific equipment.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-C-19729A(Wep) Computer, Aerial Photographic Data

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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C-8279

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